

Scheme 2. Synthesis of tetrahalide propellane precursor 22.

A final literature example of a 2-methylBCP synthesis was reported by Anderson et al. in 2024 via anionic BCP bridge functionalization.^[20] This strategy is based on the radical BCP-bridge C-H bromination reported by the MacMillan group.^[19] Hence, 1,3-BCP dicarboxylic acid **18** was transformed in four steps to the 2-brominated BCP derivative **19**. Subsequent lithiation using *t*-BuLi afforded the stabilized lithiated intermediate **20**, which could be methylated to afford **21** in 50% yield.

The methyl group is a very common substituent in pharmaceuticals, and its introduction in bioactive compounds typically results in a plethora of effects that are relevant in the drug discovery process.^[36–39] The term “magic methyl effect” has been coined to describe the sometimes dramatic improvements that can be seen upon its introduction. Hence, comparing bioactive compounds containing BCPs with their methylated BCP analogues will be of great interest. For such a purpose, the availability of synthetic methodology that introduces diversity on a methylated BCP precursor would be advantageous. Herein we report our efforts toward developing a versatile methodology for the synthesis of bridge-methylated BCPs **Xa** (Scheme 1C). The novel 2-methyl[1.1.1]propellane **23a** was deemed an ideal intermediate for this purpose, as it allows mono- and difunctionalization of its central 1,3-bond. Next to the synthesis of **23a**, an investigation of its reactivity compared to that of nonsubstituted [1.1.1]propellane **23b** was undertaken. We also report our preliminary results regarding the impact of BCP 2-methylation on physical properties such as melting point and lipophilicity.

2. Results and Discussion

2.1. Synthesis of 2-methyl[1.1.1]Propellane **23a**

The synthesis of **23a** was envisioned using the conventional lithiation procedure (Scheme 1C),^[7,28] identifying **22** as the key intermediate. This was first synthesized by a Wittig reaction between 1,3-dichloroacetone **1** and ethyl triphenyl phosphonium chloride **24**, leading to **25** (Scheme 2). It was found that the use of the chloride salt^[40] gave a much better yield compared to the reported^[41] use of the corresponding bromide salt, as it avoided the formation of allylic bromide side products (detected by MS-analysis; see Supporting Information section 3.3). Despite much experimentation, however, the yield of the dibromocyclopropanation of **25** using bromoform could not be optimized, with the conditions reported by the Baran group in his synthesis of the MOMOCH₂-substituted propellane **2** (55% yield, cf Scheme 1A)^[32] only leading to a 12% yield.

An alternative route starting from the commercially available diethyl 2-ethylidenemalonate **26** was more successful

(Scheme 2). From **26**, the corresponding dibromocyclopropane **27** was obtained in 69% yield using Baird's procedure,^[42,43] with careful workup to avoid ester saponification. This reaction could be optimized on 200 g scale, employing high-vacuum distillation to isolate **27** in 72% yield. Subsequent ester reduction using a combination of LiAlH₄ and AlCl₃, which results in the formation of AlH₃, afforded diol **28** in 49% yield (see Supporting Information section 3.4). Similar yields were reported for higher alkylated cyclopropane derivatives.^[31]

The low yield of the ester reduction was initially attributed to significant product loss during aluminum salt removal in the workup. Using LiAlH₄ as a reductant led to swift consumption of starting material at 0 °C, but the monobrominated diastereomers **29** (Scheme 3) were isolated as major side products, with the major isomer assigned as the *cis*-isomer (70:30) based on coupling constant analysis (see Supporting Information section 3.3.3). Two other side products were characterized as the fully debrominated **30** and the allene **31**. Dibromocyclopropane reduction has also been reported by Baird et al. on the nonalkylated dibromocyclopropane substrate,^[44] which they showed could be avoided by using the milder alane (AlH₃). In our hands, repeated efforts of applying alane for the selective reduction remained unsuccessful, with debromination still observed. In fact, TLC analysis revealed that debromination occurred before complete consumption of the starting material.

To prevent debromination, the use of milder metal borohydrides was explored. The use of LiBH₄ in Et₂O at room temperature resulted in incomplete conversion of the starting material, with the formation of debrominated side products **29** upon longer reaction times or applying reflux conditions. To our delight, the addition of methanol in stoichiometric amounts, known to increase the reactivity of LiBH₄,^[45] resulted in fast and complete conversion of the starting material with limited formation (< 15%) of side products, which were difficult to separate. Furthermore, the workup is more convenient compared to reactions with aluminum hydrides. On a 20-gram scale, purification with a single flash chromatographic separation resulted in 74% of the pure desired diol **28**, next to a small amount of a mixture of **28** and **29**. Efforts to completely prevent debromination using the significantly less expensive NaBH₄ resulted in incomplete conversion of the starting material and long reaction times, while still observing debromination.

Bis-deoxychlorination of diol **28** using the conventional Appel conditions^[31] surprisingly led to a complex mixture from which the desired tetrahalide **22** could not be isolated (see Supporting Information section 3.3). Given that these conditions lead to the formation of two equivalents of triphenyl phosphine oxide and require the use of restricted CCl₄ as a solvent, it was decided to explore other chlorination methodologies.

The use of Vilsmeier-Haack (POCl₃) procedures^[46] also led to complex mixtures (see Supporting Information section 3.3), and Darzens chlorination conditions (thionyl chloride/pyridine or thionyl chloride/TiCl₄)^[47] led to the formation of a mixture of the corresponding diastereomeric bicyclic spiro sulfite esters **32**. A successful reaction was achieved using a method reported by Kartika et al., employing triphosgene-pyridine.^[48,49] yielding the desired compound **22**. However, the corresponding bicyclic spiro

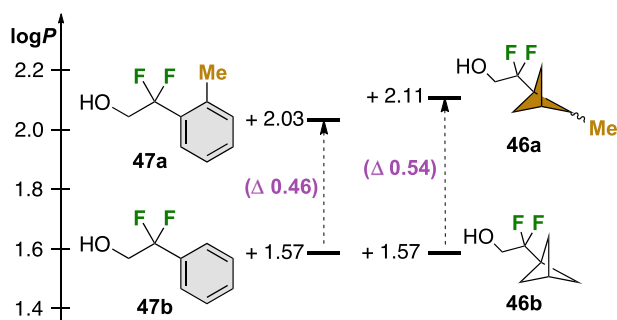


Figure 2. Comparison of measured octanol-water partition coefficients between phenyl, tolyl, BCP, and 2-MeBCP.

inseparable diastereoisomers **44** in 91% yield. Equally, oxidation of sulfide **34a** using the method of Li^[57] led to the corresponding sulfoximine diastereomers **45** without stereoselection. The Bräse group had reported a 55% yield for the oxidation of the corresponding nonmethylated phenyl BCP sulfide **34b**.^[58] Neither diastereomeric mixture **44** nor **45** could be separated using normal-phase silica gel chromatography.

2.3. Investigation of Physical Properties

The effect of the added bridge-methyl group on physicochemical properties was investigated. The lipophilicity of BCP **46b** and its aromatic equivalent **47b**, measured by a ¹⁹F NMR-based protocol developed in our group,^[59] was shown to be identical (Figure 2), while addition of a methyl group (**46a**) led to a significant increase of 0.54 logP units. This increase in lipophilicity is similar to the effect for the addition of a methyl group in *ortho*-position (**47a**, 0.46 logP unit) and what is reported in the literature,^[37] including for close linear 2,2-difluoroalkanol analogues ($\Delta \log P$ between 2,2-difluorobutanol and 2,2-difluoropentanol is 0.58 logP units).^[60] This result is in line with the increase predicted by conventional clogP methodologies (see Supporting Information section 1.7).

1,3-Disubstituted bicyclopentanes are often solid compounds. In contrast, most of the 2-Me-BCPs were obtained as liquids, probably due to the absence of symmetry leading to less favorable crystal packing. For example, whereas **37b** and **38b** were isolated as crystalline solids, their corresponding 2-Me analogues were liquids at room temperature. However, 2-methylated BCP derivatives **43a** and **49a** were found to be solids (Figure 3A/B), with melting points much lower than that of the nonmethylated equivalents, especially for **49a/49b** ($\Delta T = 87^\circ\text{C}$). The melting point of the corresponding *n*-alkyl sulfone **50** is lower than that of the BCPs **49a** and **49b**, as is the melting point of **48** compared to that of **43a/43b**. This is in agreement with the typically lower melting points of open-chain alkanes compared to cyclic structures. Such melting point reduction upon bridge functionalization can also be found upon chlorination (Figure 3C). Whereas dimethyl bicyclopentane dicarboxylate **51** has a melting point of 90–91 °C,^[61,62] this is much reduced for the monochlorinated derivative **52**.^[13] The melting point of the bridge-dichlorinated **53** is similar to that of **52**.^{41) [63]}

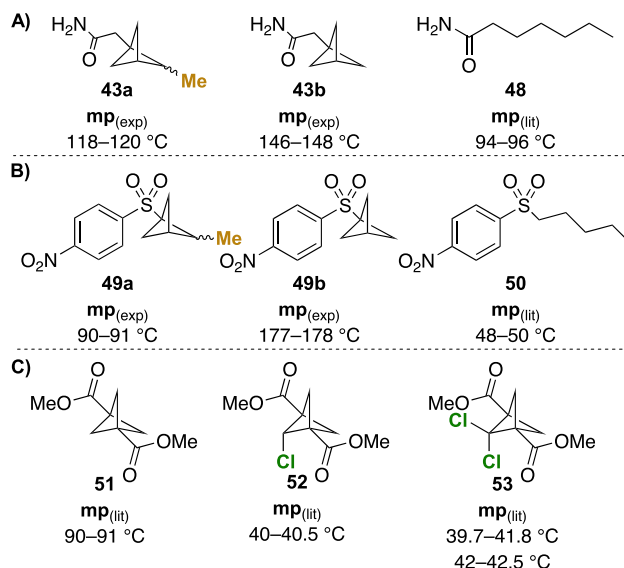
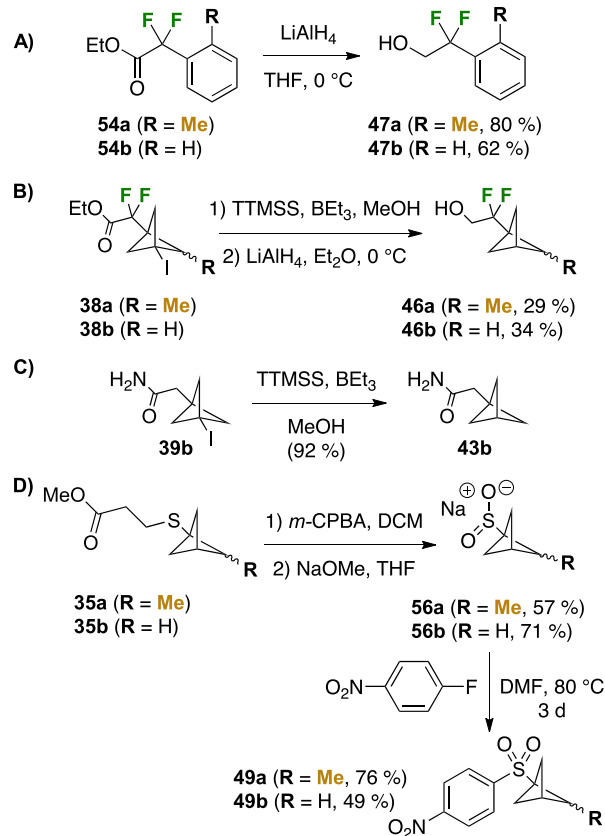


Figure 3. Melting point comparison of bridge-substituted BCPs with their nonbridge-substituted analogues.



Scheme 8. Syntheses of substrates **43a–49b**.

The syntheses of the novel substrates shown in Figures 2 and 3 are shown in Scheme 8. The aromatic logP substrates **47a** and **47b** were obtained by reduction of the corresponding ethyl esters **54a** and **54b**, respectively (Scheme 8A), which were either synthesized according to a reported method (**54a**),^[64] or were commercially available (**54b**). Substrates **46a** and **46b** were obtained by radical-mediated iodine reduction^[55] of **38a** and